



KuCoin decentralises

KuCoin launches a decentralised crypto wallet, with its app expected to launch later in June. <https://dgit.in/jun22-62>



Entra is here!

Microsoft announces Entra, a new product family of identity and access management solutions. <https://dgit.in/jun22-63>

Best Snapdragon 8 Gen 1 phone?

Snapdragon 8 Gen 1 is Qualcomm's flagship mobile chipset. But which phone handles it the best?

Sushant Talwar | sushant@dgit.in

Q

Qualcomm's been one of the biggest names in the chip-making business for years. While it looked like

the script was going to be the same for 2022 as well, in reality, Qualcomm's been facing a lot of heat for its flagship, Snapdragon 8 Gen 1 SoC.

Yes, this is the same chipset that we've seen on top-of-the-line Android phones from the likes of Samsung, Xiaomi, OnePlus and some others. And also the one which has been getting a lot of negative coverage for the insane amount of heat it's producing compared to its previous generation flagship chipsets.

Now honestly, heat isn't always a bad thing. Any chipset on a phone will create heat, which can be managed either by the chipset itself or by the phone to ensure close to optimal performance. Of the two approaches, the latter is better as heat management on the chipset's end is the last resort where the chip throttles itself to prevent permanent damage to the silicon.

But it appears with the Snapdragon 8 Gen 1, heat generated by the chipset is so high that phones are not able to effectively handle it, and in turn, the chipset is throttling itself quite aggressively

to keep the thermals in check. That does not mean all Snapdragon 8 Gen 1 phones are facing similar issues. There are phones which handle thermals relatively better than the rest.

Today, we are going to be looking at four phones – Xiaomi 12 Pro, OnePlus 10 Pro, Galaxy S22 Ultra and Realme GT 2 Pro. Which can handle the thermals of the Snapdragon 8 Gen 1 the best? We will be putting these through a series of benchmarks, stress tests, and gaming scenarios to find out which of

on the one found on the Snapdragon 888 from last year, with the X2 core being 16 per cent faster, the A710 being 10 per cent faster, and the A510 being 35 per cent faster than the comparative cores found on the previous generation chip. But with great power, comes great... heat. And from what we've seen, despite their promises, Qualcomm and Samsung have not managed to keep the heat generation on the 8 Gen 1 to acceptable levels, leading to serious chipset throttling in some cases.



these four phones delivers the most stable and powerful performance using the new monster chip from Qualcomm.

WHY IS THE SNAPDRAGON 8 GEN 1 OVERHEATING?

Before we get to how the Snapdragon 8 Gen 1 is overheating, it is important to understand why it's possibly doing so. The Snapdragon 8 Gen 1 is built using Samsung's new 4nm process and features one Arm Cortex X2 core for performance, three "big" Cortex-A710 cores, and four Cortex-A510 "little" cores for high-efficiency workloads. This architecture is a serious upgrade

WHICH SNAPDRAGON 8 GEN 1 PHONE HANDLES THE HEAT BEST?

So, now that we've talked about why the Snapdragon 8 Gen 1 creates a massive amount of heat, it's important to know which of the current crop of phones handles the chipset the best.

There are a lot of factors that play a part in effectively handling extra heat at the phone's end. The presence of a big cooling chamber as well as the use of materials that help in quick dissipation of heat are obviously important, but an OS that's light and bloat-free also helps keep the temperatures low. While all four phones we are testing